

Supplementary Material

Current-limited capability of grid-forming converters in reserve and inertia scheduling:
formulation, sign condition and validation

Junseok Lim, Sanghyuk Ji, Byungjun Lee, Sungwoo Bae

Department of Electrical Engineering, Hanyang University, Seoul, Republic of Korea

Equation, table, figure and section numbers of the main text are cited without a prefix; those of this document carry the prefix S. All results were generated with the code and data available at <https://github.com/Junseok-Lim-HY/gfm-capability-set-clearing> (see Data availability in the main text).

S1 State-of-charge gate

The band availability of Eq. (6) rises linearly from the energy floor $\underline{\sigma}^e = 0.05$ to the breakpoint σ^* . Table S1 gives the cost difference δ against the active-power bound when σ^* is varied with the floor held; at $\sigma^* = \underline{\sigma}^e$ the gate is removed. The fleet-mean availability, evaluated on the selected (least-throughput) optimum from the cleared state-of-charge trajectories, falls from 1.00 without the gate to 0.41–0.71 at the base breakpoint and 0.35–0.69 at $\sigma^* = 0.30$; being a primal quantity on a wide optimal face, it depends on the selected optimum. The cost difference δ changes by less than 0.01 percentage points across the range. The gate is retained because a formulation crediting the short-term current to an empty store would be incorrect, not because it carries a result.

Table S1: δ (%) and fleet-mean band availability against the full-band breakpoint σ^* ($e = 0.8$ pu, $\bar{P} = S$).

σ^*	IEEE 39, 0.2		IEEE 39, 0.4		RTS-24, 0.2		RTS-24, 0.4	
	δ	avail.	δ	avail.	δ	avail.	δ	avail.
0.05 (no gate)	0.96	1.00	0.34	1.00	0.68	1.00	1.08	1.00
0.10	0.96	0.78	0.34	0.64	0.68	0.56	1.08	0.59
0.20 (base)	0.96	0.71	0.34	0.54	0.68	0.41	1.08	0.47
0.30	0.96	0.69	0.34	0.51	0.68	0.35	1.08	0.43

S2 Polygon approximation and exact second-order cone

The inscribed m -gon withholds capability in the active coordinate by up to the values in Table S2, evaluated over reactive dispatches from zero to 95 % of rating. The clearing inherits much less than the pointwise worst case: solving the exact conic problem on the same variables (constraints (3)–(4) without the $\cos(\pi/m)$ factor, solved with CLARABEL through CVXPY) gives the costs in Table S3. The polygon cost exceeds the cone cost by at most 0.066 % at $m = 24$, and $\delta_m \leq \delta_{\text{exact}}$ in every case, so the reported differences are conservative.

Table S2: Worst shortfall of the inscribed polygon in the active coordinate and the radial bound $1 - \cos(\pi/m)$ (%).

m	Active-coordinate shortfall	Radial bound
8	10.76	7.61
12	6.72	3.41
24	2.70	0.86
48	0.34	0.21
96	0.15	0.05
192	0.03	0.01

Table S3: Polygon against exact cone: excess of the polygon clearing cost over the conic cost (%), and δ (%) against the active-power bound for the polygon at three face counts and for the cone.

System, share	e	Excess over cone			δ			
		$m=24$	$m=96$	$m=384$	$m=24$	$m=96$	$m=384$	cone
IEEE 39, 0.2	0.8	0.000	0.000	0.000	0.96	0.96	0.96	0.96
	1.0	0.025	0.002	0.000	9.00	9.02	9.03	9.03
IEEE 39, 0.4	0.8	0.000	0.000	0.000	0.34	0.34	0.34	0.34
	1.0	0.066	0.004	0.000	14.87	14.93	14.93	14.93
RTS-24, 0.2	0.8	0.013	0.001	0.000	0.68	0.69	0.69	0.69
	1.0	0.024	0.002	0.000	1.17	1.19	1.20	1.20
RTS-24, 0.4	0.8	0.000	0.000	0.000	1.08	1.08	1.08	1.08
	1.0	0.000	0.000	0.000	1.92	1.92	1.92	1.92

S3 Variable domains and additional constraints

$$\begin{aligned}
0 \leq p_i^{d,t} \leq P_i^{\text{sch}}, \quad 0 \leq p_i^{c,t} \leq P_i^{\text{sch}}, \quad 0 \leq c_i^t \leq e_i^t, \\
0 \leq Q_i^t, \tilde{Q}_i^t, R_i^t \leq S_i, \quad 0 \leq H_i^t \leq I^s S_i, \quad 0 \leq a_i^t \leq 1, \\
\underline{\sigma}^e \leq \sigma_i^t \leq \bar{\sigma}, \quad \underline{P}_g \leq p_g^{g,t} \leq \bar{P}_g, \quad 0 \leq r_g^{g,t} \leq \bar{R}_g, \quad p_g^{g,t} + r_g^{g,t} \leq \bar{P}_g.
\end{aligned} \tag{S1}$$

The synchronous lower bound is the minimum stable output, because Eq. (16) counts the stored energy of every committed machine. The upper bound $I^s S_i$ on H_i^t applies to the capability set; under the comparison models of Section 4.2 it is S_i (active-power bound, apparent-power model) and $\gamma^{\text{UD}} S_i$ (uniform derating), where it is never active. Under the uniform derating, $Q_i^t \leq \kappa^Q S_i$ replaces $Q_i^t \leq S_i$. Reactive output is non-negative; allowing absorption as a separately priced column leaves every reported cost, price and voltage unchanged and is never used, because every zonal reactive requirement binds. The linear relaxation of the store is tight: $\min(p_i^{d,t}, p_i^{c,t}) = 0$ for every converter and period in every reported schedule.

S4 Prices as directional derivatives

For all 1,648 requirement rows of the eight configurations (energy balance, reserve, zonal reactive and per-outage inertia rows), the cleared cost was re-evaluated at ± 1 MW of the right-hand side. Every reported multiplier lies between the two one-sided differences, as a subgradient must. The 64 inertia rows are degenerate (the two differences differ), so a single row's multiplier is not unique. The aggregate inertia price of Section 3.2, obtained by moving all inertia rows of a period together, has coinciding left and right differences at steps of 1, 0.25 and 0.01 MW in

every configuration and dispatch level (Table S4), so the reported inertia prices and their ratios are unique.

Table S4: Aggregate inertia price (\$/MW/h): reported value and one-sided differences at three perturbation sizes (identical at all three).

System, share	e	Active-power bound			Capability set		
		reported	left	right	reported	left	right
IEEE 39, 0.2	0.8	8.00	8.00	8.00	3.00	3.00	3.00
	1.0	41.00	41.00	41.00	3.00	3.00	3.00
IEEE 39, 0.4	0.8	8.00	8.00	8.00	3.00	3.00	3.00
	1.0	41.00	41.00	41.00	3.00	3.00	3.00
RTS-24, 0.2	0.8	8.00	8.00	8.00	3.003	3.003	3.003
	1.0	41.00	41.00	41.00	3.00	3.00	3.00
RTS-24, 0.4	0.8	6.75	6.75	6.75	3.00	3.00	3.00
	1.0	23.25	23.25	23.25	3.00	3.00	3.00

S5 Reduction test

With $I^s = I^c$, no reactive requirement, $V = 1.0$ pu and $m = 192$, the capability set and the active-power bound with $\bar{P} = S$ return the same inertia prices and costs within 0.0023 % (Table S5); the 192-gon can impose up to 0.027 % on the active coordinate.

Table S5: Reduction test.

System, share	Price, bound	Price, set	Cost, bound	Cost, set	Gap (%)
IEEE 39, 0.2	8.00	8.00	3383389	3383418	0.0008
IEEE 39, 0.4	8.00	8.00	2441213	2441264	0.0021
RTS-24, 0.2	8.00	8.00	1495550	1495563	0.0009
RTS-24, 0.4	6.75	6.75	867606	867626	0.0023

S6 Full sensitivity table and scale check

Table S6 lists every one-at-a-time sensitivity. Table S7 gives the IEEE 118-bus check discussed in Section 5.3: converters at $P/S = 0.95$, dc approximation, as-shipped case.

S7 Post-contingency deliverability: pre-correction screen

Table S8 gives the post-contingency screen of the lossless schedules against the continuous branch ratings for every generator and converter outage, before Eq. (21) is imposed. The pre-contingency loading is at rating in every configuration because the clearing binds a branch constraint, so most outages overload some branch; the secured formulation of Section 3.4 removes this for the secured set (Table 7 of the main text). “Short” counts outages whose loss exceeds the fleet holding, which the secured formulation covers by inertial pickup of the surviving machines.

Table S6: δ (%) against the active-power bound, one parameter varied at a time ($e = 0.8$ pu, $I^s = 1.5$ pu, $\bar{P} = S$; base case in bold).

Quantity	Value	IEEE 39		RTS-24		Quantity	Value	IEEE 39		RTS-24	
		0.2	0.4	0.2	0.4			0.2	0.4	0.2	0.4
RoCoF (Hz/s)	0.5	8.51	1.81	6.77	4.99	V offset	-0.05	0.96	0.34	0.39	1.08
	1	0.96	0.34	0.68	1.08		-0.02	0.96	0.34	0.59	1.08
	1.5	0.00	0.00	0.16	0.00		0	0.96	0.34	0.68	1.08
	2	0.00	0.00	0.16	0.00		+0.02	0.96	0.34	0.73	1.08
$\mathcal{H}_g$ (s)	3	4.62	0.91	2.63	2.08	Offer ratio	+0.05	0.96	0.34	0.74	1.08
	4	0.96	0.34	0.68	1.08		0.25	0.24	0.09	0.17	0.28
	6	0.00	0.00	0.16	0.00		0.5	0.48	0.17	0.34	0.55
	1	2.52	1.19	0.71	1.21		1	0.96	0.34	0.68	1.08
Storage (h)	2	1.80	0.89	0.71	1.28	Cycling (\$/MWh)	2	1.86	0.65	1.36	2.08
	4	0.96	0.34	0.68	1.08		4	3.53	1.22	2.62	3.83
	8	0.56	0.00	0.50	0.57		0	0.96	0.34	0.68	1.08
	8	0.96	0.34	0.27	1.08		1	0.95	0.34	0.69	1.08
Faces m	12	0.96	0.34	0.55	1.08	P/S	2	0.95	0.33	0.69	1.07
	24	0.96	0.34	0.68	1.08		5	0.95	0.33	0.68	0.99
	48	0.96	0.34	0.69	1.08		1.00	1.02	0.53	0.62	5.44
	96	0.96	0.34	0.69	1.08		0.95	0.98	0.41	0.67	2.67
Overlap χ	192	0.96	0.34	0.69	1.08	$\bar{P} = P^{\text{sch}}$	0.90	0.93	0.26	0.56	1.07
	384	0.96	0.34	0.69	1.08		0.85	0.87	0.07	0.27	0.28
	0	0.96	0.34	0.68	1.08			3.67	1.90	1.40	7.94
	0.25	0.96	0.34	0.68	1.08						
	0.5	0.96	0.34	0.68	1.08						
	1	0.96	0.34	0.68	1.08						

S8 RMS converter model

States. Angle ϑ of the internal source against the Thévenin source, per-unit angular frequency ω of the virtual rotor, internal source magnitude E^v (the symbols δ and E_i of the main text are not used in this section).

Network (algebraic). With $\bar{E} = E^v e^{j\vartheta}$, filter reactance x^f and reactance x^g to a source of magnitude $V^g(t)$ at angle zero,

$$\bar{I} = \frac{\bar{E} - V^g}{j(x^f + x^g)}, \quad \bar{V} = \bar{E} - jx^f \bar{I}, \quad P + jQ = \bar{V} \bar{I}^*, \quad (\text{S2})$$

with axis currents $i_d = P/|\bar{V}|$ and $i_q = Q/|\bar{V}|$.

Limiter. The available magnitude is I^c outside the short-term band and I^s inside it (the band opens at the event and lasts T^b). When $\|(i_d, i_q)\| > I^{\text{lim}}$ the priority axis is clipped to I^{lim} and the other axis receives $\sqrt{(I^{\text{lim}})^2 - i_{\text{kept}}^2}$. The delivered powers are $P^{\text{out}} = i_d^{\text{lim}} |\bar{V}|$ and $Q^{\text{out}} = i_q^{\text{lim}} |\bar{V}|$.

Dynamics. With grid frequency $\omega^g(t)$ falling at the design RoCoF for the containment window,

$$\dot{\vartheta} = \omega_{\text{base}}(\omega - \omega^g(t)), \quad (\text{S3})$$

$$2H^v \dot{\omega} = P^{\text{ref}}(t) - P^{\text{out}} - D(\omega - \omega^g(t)), \quad (\text{S4})$$

$$\dot{E}^v = K^q(Q^{\text{tgt}} - Q^{\text{out}}), \quad (\text{S5})$$

where P^{ref} is P before the event and $P + \chi R$ after it, $H^v = H f_0 / (2 \text{RoCoF})$ so that the scheduled H is requested at the design RoCoF, and Q^{tgt} is the scheduled Q (or $Q + (V^{\text{pre}} - |\bar{V}|)/k^v$ in the droop variant).

Table S7: IEEE 118-bus check ($P/S = 0.95$, dc approximation).

Share	RoCoF (Hz/s)	e (pu)	δ (%)	Price, bound	Price, set	Binding
0.2	1.0	0.8	0.00	0.00	0.00	no
0.2	1.0	1.0	-0.68	0.00	0.00	no
0.2	0.5	0.8	0.00	0.00	0.00	no
0.2	0.5	1.0	-0.68	0.00	0.00	no
0.4	1.0	0.8	0.00	0.00	0.00	no
0.4	1.0	1.0	-0.68	0.00	0.00	no
0.4	0.5	0.8	0.00	3.00	3.00	yes
0.4	0.5	1.0	+0.12	8.00	3.00	yes

Table S8: Post-contingency screen of the lossless schedules against continuous ratings (all outages).

System, share	Model	Outages	Over rating	Worst (%)	Short
IEEE 39, 0.2	bound	72	64	158.1	14
IEEE 39, 0.2	set	70	64	161.2	16
IEEE 39, 0.4	bound	73	54	152.6	1
IEEE 39, 0.4	set	70	62	153.6	0
RTS-24, 0.2	bound	224	200	121.2	3
RTS-24, 0.2	set	251	142	105.1	1
RTS-24, 0.4	bound	221	177	132.1	1
RTS-24, 0.4	set	221	177	138.1	0

Anti-windup. When the limit binds the angle is held at

$$\vartheta^{\max}(t) = \arccos \frac{(E^v)^2 + (V^g)^2 - [(x^f + x^g)I^{\lim}]^2}{2E^vV^g}, \quad (\text{S6})$$

with $\dot{\omega} \leq 0$ there, and $\dot{E}^v \leq 0$ while the current is saturated.

Initialisation and numerics. The terminal voltage is held at the scheduled value and the sources follow; fixed-step fourth-order Runge–Kutta with a step of 10^{-4} s. Parameters: $f_0 = 60$ Hz, $x^f = 0.05$ pu, $D = 20$, $K^q = 20 \text{ s}^{-1}$, $k^v = 0.04$ pu, event at 0.5 s, containment window and band 1 s, $I^c = 1.0$ pu, $\chi = 1$, $P = 0.60$, $R = 0.10$, low and high reactive schedules $Q = 0.20$ and 0.60 (pu of rating). The model is one converter against a Thévenin source, without inner loops, dc-link dynamics, phase-locked loop, fault current or protection; it does not replace an electromagnetic-transient study.

S9 Crossing values of the Proposition

Table S9 lists the crossing $\tilde{Q}^{\text{crit}}$ of the Proposition (Section 2.5) that Fig. 3 of the main text shows as a map, for $\bar{P} = S$ and full band availability.

Table S9: Crossing $\tilde{Q}^{\text{crit}}$ (pu of rating) at $\bar{P} = S$ and full band availability. A dash indicates $L \leq \bar{P}$.

V (pu)	I^s (pu)				
	1.00	1.10	1.20	1.35	1.50
0.95	—	0.303	0.547	0.803	1.015
0.98	—	0.403	0.619	0.866	1.077
1.00	—	0.458	0.663	0.907	1.118
1.02	0.201	0.509	0.706	0.947	1.158
1.05	0.320	0.578	0.767	1.005	1.217

S10 Source-case corrections and parameters

Table S10 lists the voltage controls moved to bring the two source cases within their declared voltage limits (Section 4.1); active power is not redispatched. Table S11 lists the parameters of the case study (Section 4.2).

Table S10: Voltage controls adjusted to bring the source cases within their declared voltage limits.

System	Control	Bus	Before	After
IEEE 39	generator setpoint (pu)	35	1.0636	1.0586
RTS-24	generator setpoint (pu)	13	0.9800	0.9950
RTS-24	generator setpoint (pu)	14	1.0140	1.0240
RTS-24	generator setpoint (pu)	17	1.0500	1.0450
RTS-24	generator setpoint (pu)	20	1.0500	1.0400
RTS-24	generator setpoint (pu)	22	1.0500	1.0550
RTS-24	transformer tap position	23	1	−1
RTS-24	transformer tap position	10	1, 1	−2, −1
RTS-24	transformer tap position	11	1, 1	2, 3

Table S11: Parameters (offers in \$/MWh for energy and cycling, \$/MW/h for availability, \$/MVar/h for reactive power).

Quantity	Value
Periods T , length Δ	8, 3 h
Reserve requirement ρ	6 % of demand
Reactive requirement φ	35 % of zonal reactive load
RoCoF limit $\dot{f}^{\max}$	1.0 Hz s ⁻¹
Synchronous inertia constant $\mathcal{H}_g$	4 s
Current limits I^c, I^s	1.0 pu, 1.5 pu
Reactive priority α_i , overlap χ	1.0, 1.0
Storage duration, initial σ	4 h, 0.60
Efficiencies η_c, η_d	0.95, 0.95
$\underline{\sigma}^e, \sigma^*, \bar{\sigma}$	0.05, 0.20, 0.95
Delivery windows τ^R, τ^H	30 min, 2 s
Polygon faces m	24
Converter offers $\kappa^e, \kappa^r, \kappa^h, \kappa^q$	8, 4, 3, 0.01
Synchronous offers κ^{ge}, κ^{gr}	38, 9
Synchronous reserve limit $\bar{R}_g$, minimum output $\underline{P}_g$	0.15 $\bar{P}_g$, case data
Tie-break cost slack (Section 3.3, step 1)	10 ⁻⁹ (relative)
Available converter injection e_i	0.8 pu of S_i